LINEAR Regression → 3 ⇒

→ Regularization, NN

① Intuition

② Recap

③ Agenda → code from scratch (GD)

→ Performance of ML model → R^2 → adj R^2

→ Statsmodel → LR ⇒ Statistics ② Imp

→ Assumptions Linear regression

Recap ⇒

$\hat{y} = w^T x + w_0$

MSE → loss func.

$L = \frac{1}{n}\sum_{i=1}^{n}(y^{(i)} - \hat{y}^{(i)})^2$ → Loss ⇒ MSE

$\underset{w,w_0}{\text{argmin}} \frac{1}{n}\sum_{i=1}^{n}(y^i - \hat{y}^i)^2$ → min.] optimisation

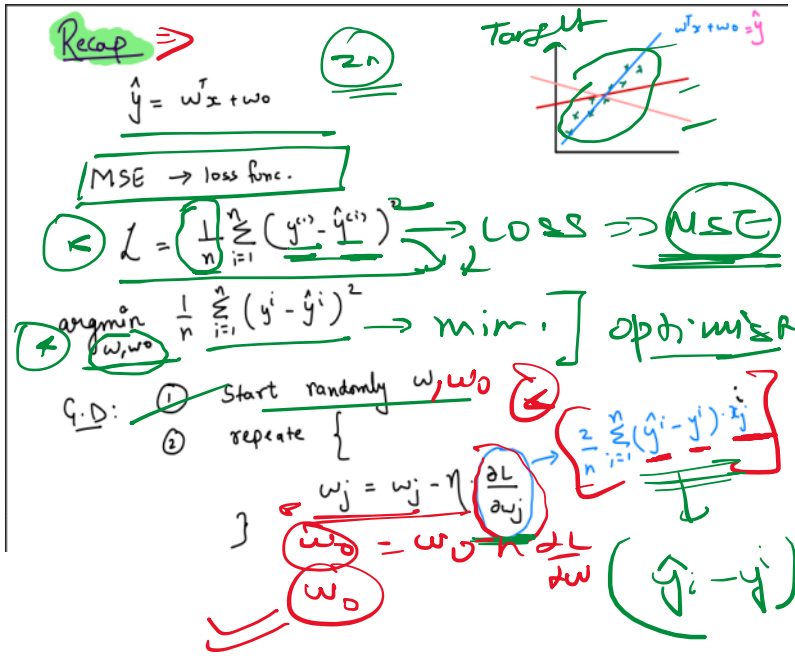
G.D: ① Start randomly w, w_0

② repeat {

$w_j = w_j - \eta \left[\frac{\partial L}{\partial w_j}\right]$ → $\frac{2}{n}\sum_{i=1}^{n}(\hat{y}^i - y^i)\cdot x_j^i$

$w_0 = w_0 - \eta \frac{\partial L}{\partial w}$ → $(\hat{y}_i - y_i)$

}



→ Best fit Line prediction

→ $w^T x + w_0 = \hat{y}$

2D, 3D ---- nD



$X \cdot w$ → pred^n → $\hat{y}$

Model score

good or Bad line ⇒ MSE or Loss

⇒ Challenges with MSE

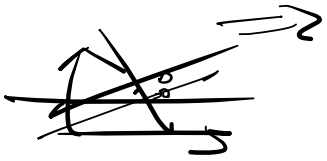
=> Challenges with RSS

↳ unbounded

ERROR

↳ (14) (13) (3000)

(1 Lac)



idea → R^2 R² - score.

coeff. of determination

CONTINUOUS

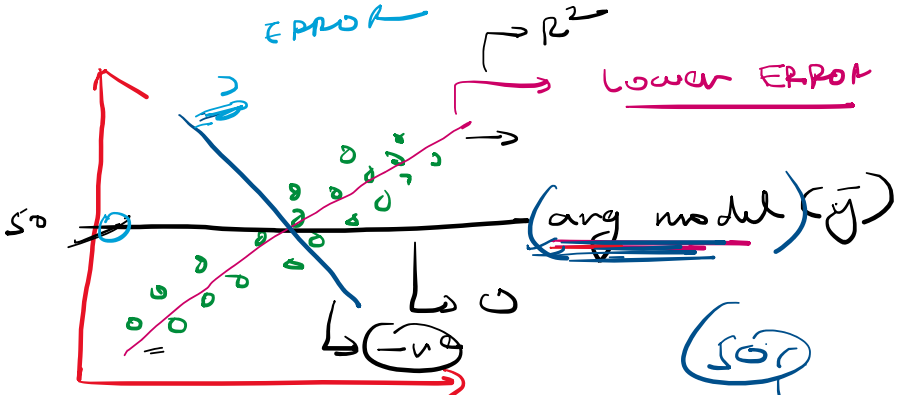
Accuracy ✗ ⇒

② Regressor →

↳ ② Performance or R sq value

① Rsquare ⇒

Dumb model → avg model



Lower ERROR (Higher performance)

↳ How much better is this from avg model

Engine

SSR

mean avg model

R^2

R2 score

700 cc → avg price

1200 cc → avg price

1800 cc → avg price

SS residual → [MSE of good model]

$R2 - SCORE = 1 - \frac{SS_{model}}{SS_{total}}$

SS_{total} → [MSE of avg model]



PRED

$\hat{y} = w_1 x + w_0$

$$R_2 - SCORE = 1 - \frac{\frac{1}{n}\sum_{i=1}^{n}(y^i - \hat{y}^i)^2}{\frac{1}{n}\sum_{i=1}^{n}(y^i - \bar{y})^2}$$

model score

$y_i - \bar{y}$

ERROR

R2 score

R^2 → Highest possible value

0 - 1

$-\infty$ to 1

Dumb → Lt → mean

R2 — score

1

$R^2 = 1$

$MSE_{model} = 0$

↳ perfect model

$R^2 = 0$

Num = denom

→ pred^n → avg pred^n

→ same as avg model

$R^2 = -1$

MSE 7 avg error

→ worse than mean model

$R^2 = -ve$ → model building wrong

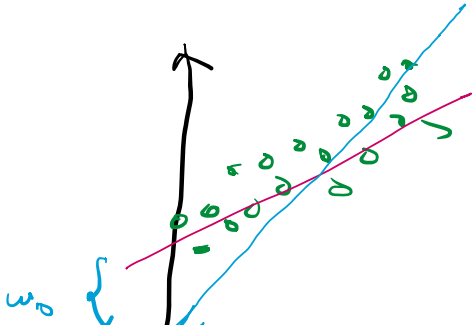
$R^2 = -ve$ →

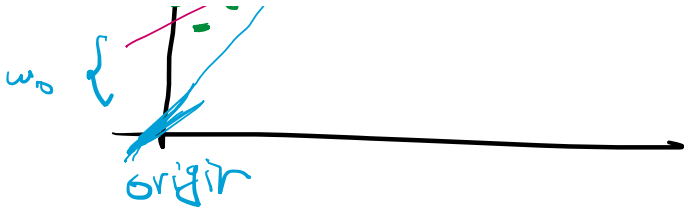
w_0 ✗

$-w_0$ R^2

$w_0 = 0$ → forced to pass through origin

w_0 ✗





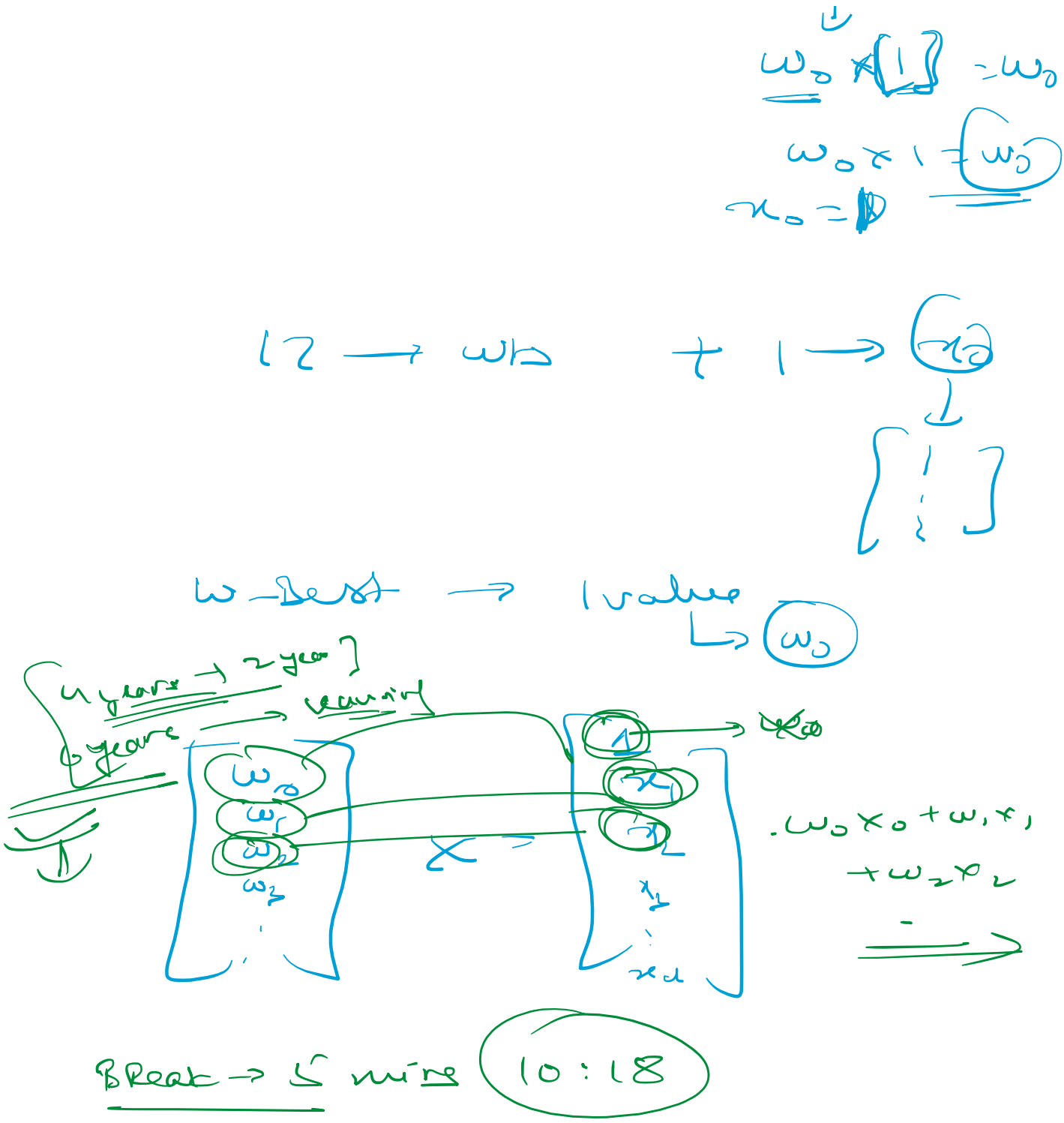
w_0

$w_1 x_1 + w_2 x_2 + \cdots + w_0$

$w_0 \cdot 1$

$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots$

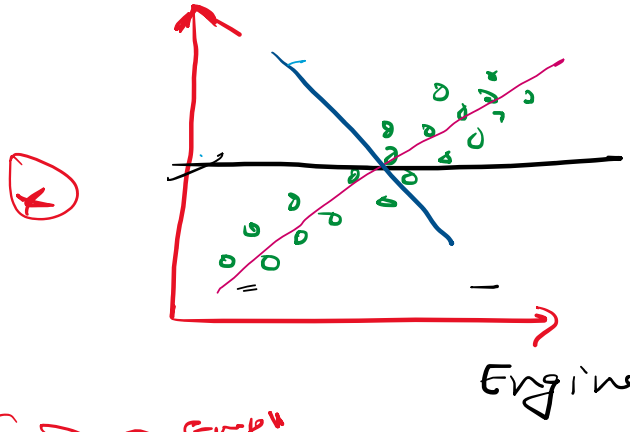
w_0

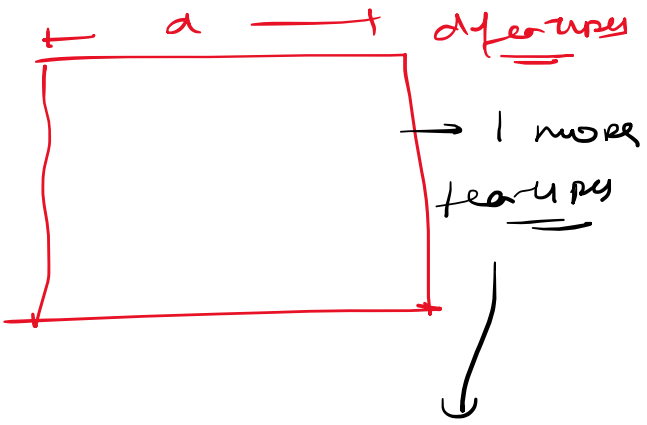


Break → 5 mins 10:18

Adj R² ⇒ Adj R2 - score

$$R^2 = 1 - \frac{SS_{model}}{SS_{total}}$$





2 cases

FEATURE relevant

$R^2 \uparrow$

FEATURE irrelevant

$R^2 \uparrow$

$R^2 \downarrow$

overfit

$R^2 \uparrow$ 0.02

0.88 → 0.9

$R^2 \uparrow$ <0

$R^2 \downarrow$

$R^2 \downarrow$

↑ or remain same

↓

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d + w_{d+1} x_{d+1}$$ new

$\Rightarrow$ Irrelevant

Correlation $\rightarrow$ No cause

≈ 0

0.02

0.03

R^2 $\uparrow$ $\Rightarrow$ slightly

No impact on prob

$\uparrow$

R^2 $\downarrow$

Adjusted R^2

Adj R^2 $= 1 - \left[\dfrac{(1-R)^2 (n-1)}{n-d-1}\right]$

$n \rightarrow$ no. of DP.

$d \rightarrow$ no. of features

d → no of features

R^2 → R2 score R^2 ↑

(1) (d+1) irrelevant → (R^2 ↑ 0.01 or 0.02)

Adj R^2 will ↓

d ↑ ⇒ (n − d − 1) ↓

$\frac{4}{2} \rightarrow \frac{4}{1}$ ↑

ENTIRE fraction ⇒ ↑

4 — ENTIRE fraction

Adj R^2 ↓

R^2 → slight

R^2 & Adj R^2

R^2 ↑ Adj ↓

(2) (d+1) is Relevant

Adj R^2 should also ↑

R^2 of significant amount

0.84 → 0.92 (0.9)

R^2 ↑ even if

$(1 - R^2)$ ↓

$\left(\frac{1}{\text{overall}}\right)$ with ↑ fraction

Net effect ⇒ Adj R^2 ↑ ✓

R^2 to be adjusted in a way that

Unnecessary features → Reduce score

Relevant features → ↑ the score

Quiz time!

Time Left: 7s

How does the adjusted R2 score differ from the regular R2 score?

42 users have participated

A The adjusted R2 score accounts for the number of predictors in the model. 71%

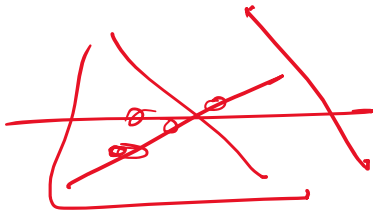
B The adjusted R2 score is always higher than the regular R2 score. 10%

predictors

good bad

A	The adjusted R2 score accounts for the number of predictors in the model.	71%
B	The adjusted R2 score is always higher than the regular R2 score.	10%
C	The adjusted R2 score considers only the explanatory power of the model.	19%
D	The adjusted R2 score is not influenced by the sample size.	0%

End Quiz Now



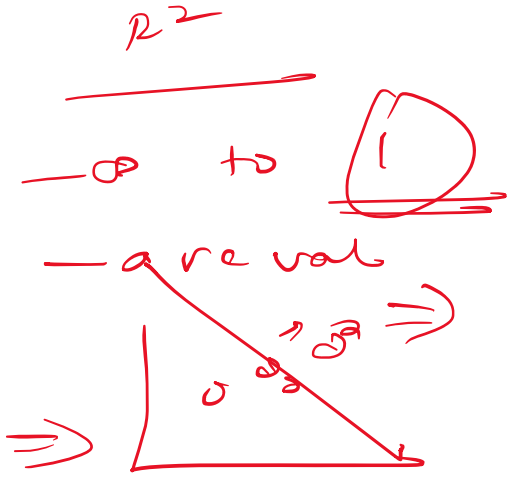
Quiz time!

Quiz Ended!

In adjusted R-squared, what is the range of possible values?

45 users have participated

A	0 to 1	27%
B	-infinity to 1	38%
C	-1 to 1	27%
D	-infinity to infinity	9%



Quiz time!

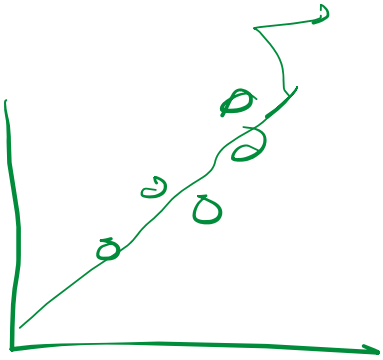
Time Left: 17s

What does a higher Adjusted R-squared value indicate about the regression model?

37 users have participated

A	The model is a perfect fit to the data.	22%
B	The model is overfitting the data.	14%
C	The model explains more variance in the dependent variable.	62%
D	The model has high bias.	3%

End Quiz Now



Statsmodel

not really used

Statistics

Maths

Stats

LR

(Disadvantages)

$$w = (X^T X)^{-1} \cdot (X^T y)$$

GD

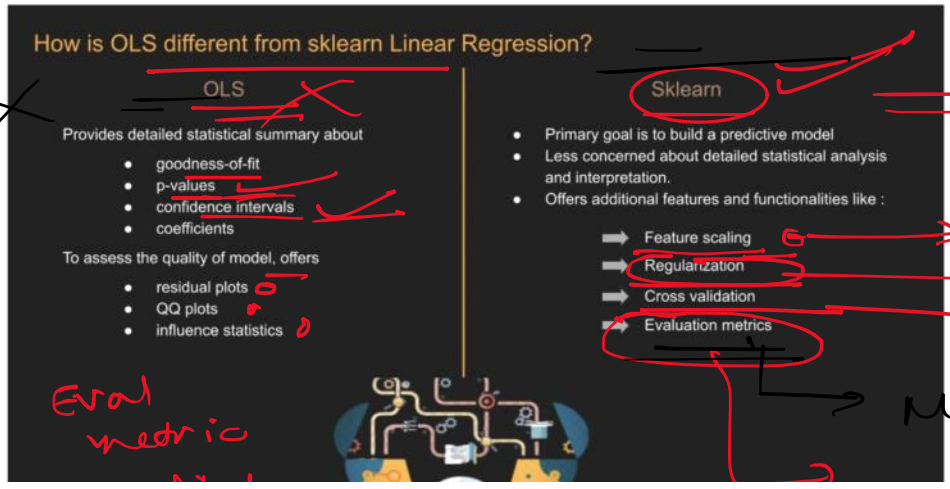
Gradient

Closed form

eg: Normal

OLS → ORDINARY Least Squares

[weights → w_1, w_2] ⟹



Better algos

Eval metric



MSE, r2 score

model.fit(x,y)

Scratch → statsmodels → R^2

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.945
Model:                            OLS   Adj. R-squared:                  0.945
Method:                 Least Squares   F-statistic:                 1.701e+04
Date:                Fri, 08 May 2026   Prob (F-statistic):               0.00
Time:                        17:21:57   Log-Likelihood:                -24523.
No. Observations:               15856   AIC:                         4.908e+04
Df Residuals:                   15839   BIC:                         4.921e+04
Df Model:                          16
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const          -8.84e-05   1.9e-06    -46.524      0.000   -9.21e-05   -8.47e-05
year               0.0015   8.92e-05     16.357      0.000       0.001       0.002
km_driven      -1.487e-06   1.82e-07     -8.150      0.000   -1.84e-06   -1.13e-06
mileage            -0.0442      0.004    -11.846      0.000      -0.051      -0.037
engine              0.0002   4.37e-05      4.547      0.000       0.000       0.000
max_power           0.0019      0.000      4.114      0.000       0.001       0.003
age                -0.1803      0.004    -46.512      0.000      -0.188      -0.173
make                0.0688      0.005     15.187      0.000       0.060       0.078
model               0.8602      0.005    179.756      0.000       0.851       0.870
Individual         -0.1406      0.019     -7.290      0.000      -0.178      -0.103
Trustmark Dealer   -0.1252      0.092     -1.359      0.174      -0.306       0.055
Diesel              0.1145      0.075      1.526      0.127      -0.033       0.262
Electric            2.9648      0.476      6.222      0.000       2.031       3.899
LPG                 0.2877      0.173      1.658      0.097      -0.052       0.628
Petrol             -0.1737      0.077     -2.249      0.025      -0.325      -0.022
Manual             -0.0914      0.030     -3.007      0.003      -0.151      -0.032
5                  -0.2927      0.083     -3.532      0.000      -0.455      -0.130
>5                 -0.4830      0.090     -5.361      0.000      -0.660      -0.306
==============================================================================
Omnibus:                     3218.068   Durbin-Watson:                   2.002
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            80809.770
Skew:                           0.347   Prob(JB):                         0.00
Kurtosis:                      14.038   Cond. No.                     1.86e+21
==============================================================================
```

p-value

conf. interval

OLS → X

p-val → 0.05

discard

→ Recursive feature elimination

feature imp —

→ X —

Prob Stats → Cond^ns

True →

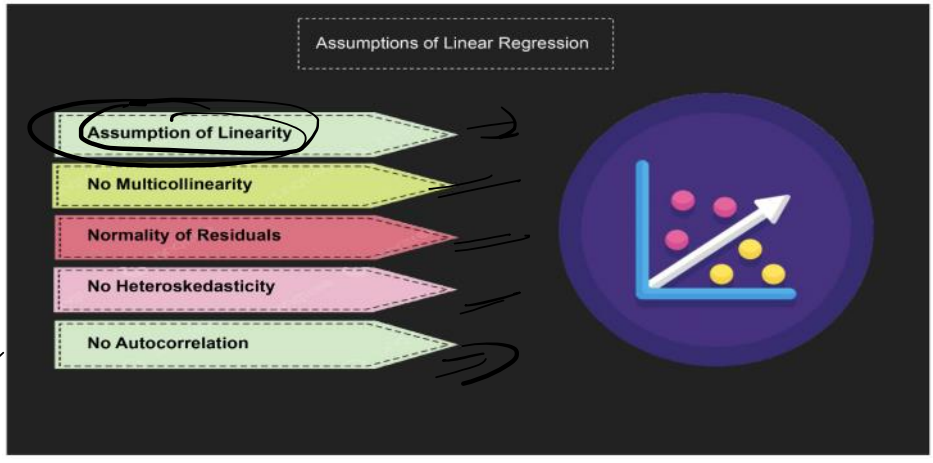
True →

True
True
True
True

Apply → LR

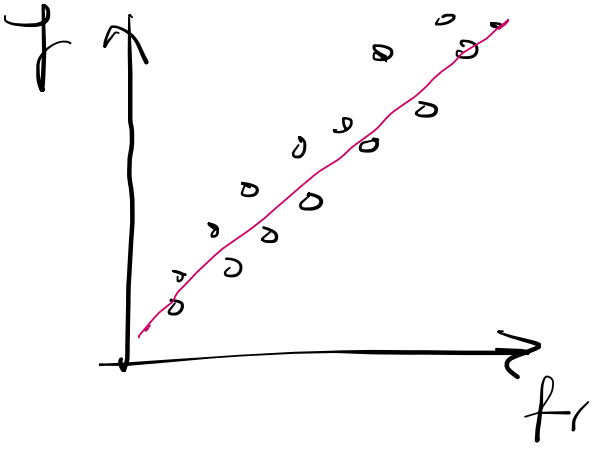
Assumptions LR ⇒ LR

⇒ Interview



LR

Linearity

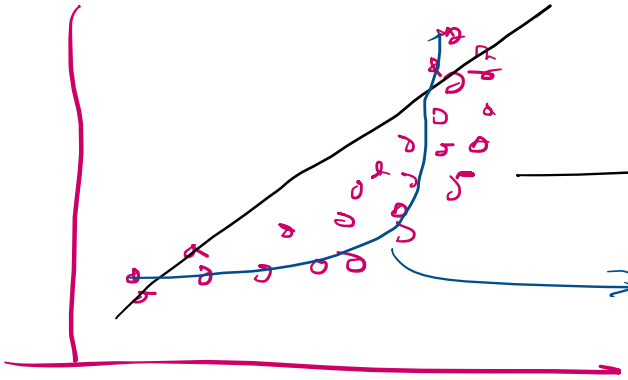


⇒ Lin eq^n

2D

Linear models

MSE



Data not Linear

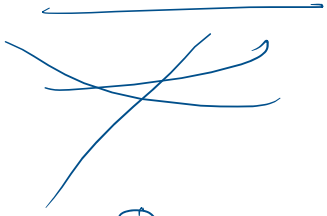
Non Linear → x^2

Linear model →

Poly nomial

Non Linear models →

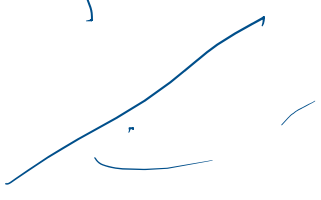
⇒ NN



x_0 x_1 x_2 x_3

x_1 VS target

Plot



RMS

LR → single model

Calculate the loss value

Given the data **x** (x[i] represents the values of independent variables for each ith observation), **y** (y[i] represents the value of the dependent variable for ith observation), and **w** (weights), return the sum of loss values for all the observations in x using the following formula:

$\frac{1}{2n}\sum_{i=1}^{n}(y_i - w^t x_i)^2$

Note: In this particular question consider the value of intercept(w0) to be zero. The loss should be rounded off to two decimal places.

Input Format:

```
Three numpy arrays of Independent Variables, Dependent Variables, and Weights are taken as an input
```

Output Format:

```
Return the loss value of the model rounded off to two decimal places.
```

Sample Input:

```
x = [[1.0, 0.0], [1.0, 1.0], [1.0, 2.0], [1.0, 3.0], [1.0, 4.0], [1.0, 5.0], [1.0, 6.0], [1.0, 7.0], [1.0, 8.0], [1.0, 9.0]]
y = [32.69, 30.56, 32.04, 32.27, 35.98, 34.39, 40.25, 39.24, 37.14, 39.38]
w = [30.0, 1.0]
```

loss

Sample Output:

```
1.81
```

Output Explanation:

```
1.81 is the loss calculated by finding sqaure of the difference between each of the y(i)
and the predicted value of the model. This loss is summed across each of the n sample
and then divided by 2n to get the mean
```

Label

MSE

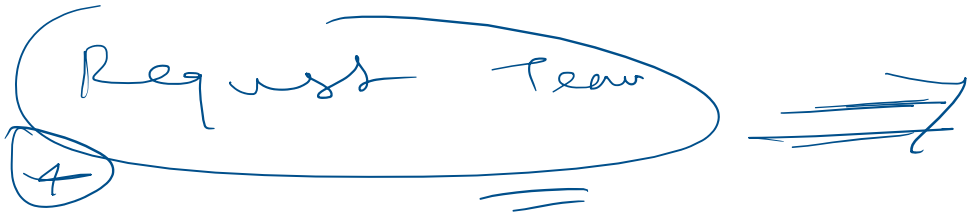
$w = [30, 1]$ Pred = $\hat{y}$

$\hat{y} = 30.1 + 1.2x$

sessions →

→ Mc

new → se


Request Fees
4